

Agent-to-Agent Protocol Enhanced Multi-Agent Coordination for Autonomous Service Provisioning in Agentic Optical Networks

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Abstract *The A2A protocol is implemented into a multi-agent system for autonomous service provisioning in optical networks. The test results demonstrate reliable end-to-end execution across heterogeneous agents, with 100% routing accuracy and low overhead, supporting interoperable and scalable optical network O&M. ©2026 The Author(s)*

Introduction

Next-generation optical networks are increasingly viewed as agentic systems, meaning that operation and maintenance (O&M) can run autonomously within an AI agent execution loop [1-4]. In this loop, an agent can not only response and reason, but also formulate plans, invoke tools, execute multi-step operations, and iteratively verify results. Recent studies have integrated multi-agent systems (MAS) into optical networks [5-9], presenting a proof of concept for lifecycle management, optical power adjustment, device configuration, performance estimation and optimization, network planning and control. Fundamentally, optical network O&M is task-driven, wherein various long-running tasks are continuously triggered by demands such as service provisioning, alarm events, performance degradation, and resource reconfiguration. These tasks usually require collaboration, interaction, and communication among multiple agents. Therefore, the key challenge in agentic optical networks lies not only in making local decisions, but also in reliably orchestrating and executing end-to-end task workflows in heterogeneous network environments.

However, for optical networks spanning wide geographic areas and involving multi-vendor [10], multi-domain [11], and multi-layer devices, reliable agent communication and collaboration remain challenging [12]. Existing multi-agent frameworks still rely heavily on proprietary interfaces, framework-specific coupling, or customized integration, which limits the standardization and scalability of multi-agent collaboration. In this context, the Agent-to-Agent (A2A) protocol [13] provides an open and vendor-agnostic way to support agent discovery, task lifecycle coordination, and cross-system interoperability, making it a promising solution for long-running multi-agent workflows in heterogeneous optical networks. Recent studies have considered A2A for data communication and computer networks [14,15], with the A2A-T extension further explored for

telecom scenarios [16]. However, the application of A2A in optical networks remains unexplored, and future efforts are required to enable its deployment with specific validation.

In this paper, we implement A2A protocol into a multi-agent coordination framework for agentic optical networks. The autonomous service provisioning is realized as a representative long-running O&M task. The proposed scheme supports unified capability exposure, task decomposition, inter-agent routing, asynchronous state tracking, and result aggregation. The test results show that A2A enables reliable end-to-end collaboration among heterogeneous agents, achieving 100% agent routing accuracy with low routing overhead.

A2A protocol-enhanced multi-agent framework for agentic optical networks

The A2A protocol provides a unified communication and collaboration mechanism for AI agents developed on different platforms. Based on the Google A2A protocol [13], we propose a multi-agent coordination framework for agentic optical networks, which defines AgentCards, specifies capability boundaries, and supports structured outputs, as shown in Fig. 1.

The proposed framework consists of a Host Agent (HA), an A2A transport layer, and multiple Remote Agents (RAs). The HA serves as the workflow entry point and interacts with the operator, while the RAs provide domain-specific functions, including strategy generation, alarm analysis and root-cause tracing, digital twin (DT)-based simulation and validation, and device configuration adjustment. Both the HA and the RAs integrate A2A interfaces for standardized communication. Although the RAs may be developed by different vendors on diverse platforms, they can expose their capabilities through a unified AgentCard [17], as illustrated in Fig. 1(b).

We design a dedicated AgentCard for each RA. For instance, in Fig. 1(b), the AgentCard of the Strategy Agent includes its basic information,

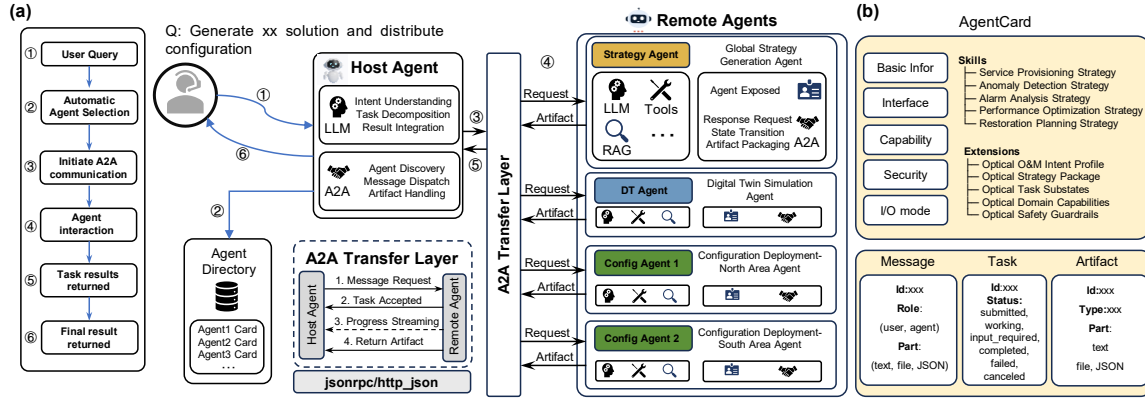


Fig. 1: (a) A2A-based multi-agent framework for autonomous optical network O&M. (b) AgentCard, Message, Task, and Artifact for agent discovery, asynchronous collaboration, and structured result exchange.

interface, exposed address, capabilities, security settings, input/output modes, and strategy-generation skills. In addition, the proposed framework extends A2A collaboration from simple message exchange to structured workflow coordination. Message, Task, and Artifact are the core interaction objects: Message carries user requests or structured inputs, Task records execution states, and Artifact returns results that can be directly reused by downstream agents. By combining task-state tracking with structured artifact exchange, the framework supports asynchronous multi-stage collaboration and intermediate-result reuse across agents, making it more suitable for long-running optical network O&M tasks.

The overall workflow for a task request is shown in Fig. 1(a). First, the operator submits an O&M request to the HA, which parses the intent and performs task orchestration. Second, the HA queries the agent directory to identify the RA whose capability best matches the requested task. Third, after selecting the target RA, the HA initiates communication through the A2A transport layer by sending a Message that contains text, files, or JSON content, as shown in Fig. 1(b). Fourth, after receiving the Message, the selected RA creates a Task and returns its creation result to the HA. Fifth, the interaction enters an asynchronous execution loop, in which the RA performs task-level execution, interacts with the control and physical layers, and continuously reports task-state updates so that the HA can monitor the progress in real time. During this process, intermediate and final result are returned as Artifacts for direct reuse by subsequent agents. Finally, after the workflow is completed, the HA organizes the task results and returns them to the operator. In this way, the proposed system goes beyond simple message passing and forms a closed, task-driven coordination loop between the HA and heterogeneous RAs. By using a standardized A2A transport interface based on

JSON-RPC and HTTP, it decouples orchestration from domain execution and supports scalable integration of heterogeneous agents in optical network O&M workflows.

Scenario-Driven Validation and Discussion

To demonstrate the proposed A2A-enhanced multi-agent framework, we consider service provisioning as a representative long-running O&M task in optical networks, where a new optical service is established through route planning, transmission feasibility verification, and device configuration delivery. The German network topology is used as the test scenario. Both the HA and the RAs are implemented using the Google Agent Development Kit (ADK) framework [18], while A2A serves as the communication layer for inter-agent collaboration. Both the HA and the four RAs are built upon an open-source foundation model family from Alibaba—Qwen3.5 [19]. Among the RAs, the Strategy Agent generates candidate working and protection paths from the source and destination nodes, and the DT Agent verifies the candidate plan using the GNPY-based network DT platform [20] and returns the corresponding device configuration parameters. To support region-level execution, the network is further divided into northern and southern areas, and the Configuration Agent is correspondingly instantiated as two regional agents, namely North Agent and South Agent. Each one is responsible for executing configuration distribution within the nodes it manages, and the specific node allocation is specified by its AgentCard.

We randomly select 10 pairs of source and destination nodes in the German network topology to perform the service provision task. Based on the operator's prompt, the HA automatically discovers the required agents and organizes the end-to-end provisioning workflow shown in Fig. 2, which consists of three stages: routing strategy

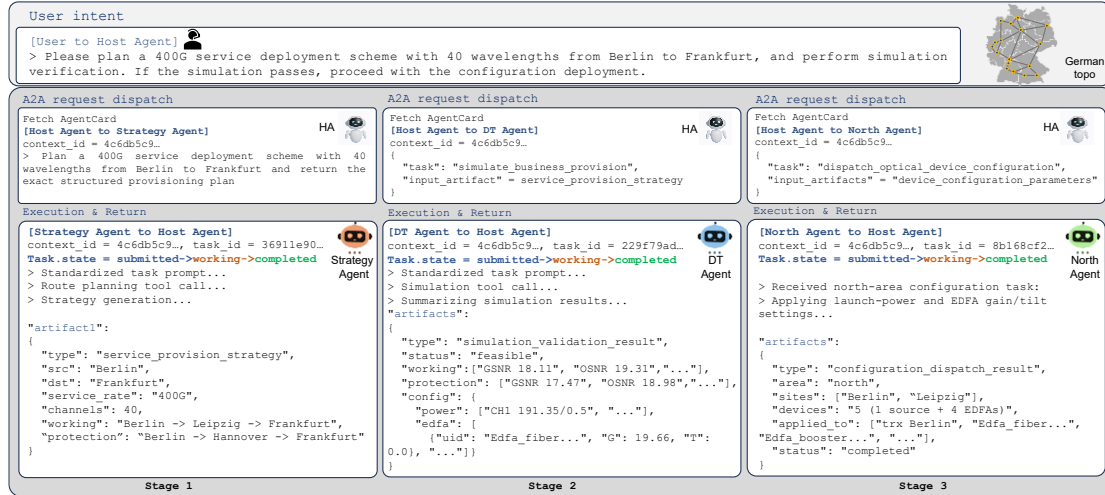


Fig. 2: HA-RA interaction process of the A2A-based MAS for executing service provisioning tasks.



Fig. 3: End-to-end workflow execution trace for 10 service provisioning tasks in the all-ADK setting.

generation, DT-assisted QoT verification, and regional configuration delivery.

As shown in Fig. 2, this task is not executed as a single response, but as an A2A-based multi-agent workflow. After receiving the operator intent, the HA first discovers the required RAs through AgentCard-based capability matching, and then dispatches the task to the Strategy Agent, DT Agent, and Configuration Agent in sequence. The representative interaction trace shows that each stage is accompanied by explicit task-state transitions, including submitted, working, and completed, together with structured artifacts returned to the HA. Specifically, the HA sends the working and protection paths generated by the Strategy Agent to the DT Agent for simulation. The DT Agent returns the verification results and device-level configuration previews. The HA then selects the corresponding Configuration Agent according to the region of the adjusted devices and delivers the device parameters, thereby completing the task loop.

Figure 3 further shows the end-to-end execution traces of 10 service provisioning tasks with detail. All tasks were completed successfully, and the routing paths vary with the configured regional scope. Compared with the time consumed by agent execution, the A2A overhead from the

Tab. 1: Impact of heterogeneous remote-agent configurations on service provisioning performance.

Setting	Success Task/10	Routing Acc.	Avg. Task Time (s)	Avg./Max A2A Overhead (s)
All ADK	10/10	100%	38.106	0.078/0.120
All LangGraph	10/10	100%	177.563	0.074/0.113
Hybrid	10/10	100%	83.768	0.068/0.098

HA to the RAs remains low, indicating that the total execution time is mainly determined by task reasoning and tool-assisted execution, while the protocol-level routing overhead is negligible.

In addition, to emulate agent heterogeneity in multi-vendor optical network scenarios, we implemented the RAs using ADK, LangGraph, and hybrid combination, and compared the overall performance of the three deployment settings in Table I. The results show that all framework combinations achieve 100% task completion and routing accuracy. Meanwhile, the average A2A overhead stays below 80 ms, and the maximum overhead remains within 120 ms in all settings. This indicates that the proposed A2A-based orchestration mechanism remains reliable and interoperable under both homogeneous and heterogeneous agent implementations, while framework heterogeneity does not introduce much extra communication cost.

Conclusions

This paper demonstrated that A2A protocol can serve as a practical foundation for multi-agent coordination in optical networks. The proposed framework supported agent discovery, task orchestration, asynchronous task tracking, and structured result exchange across each agent. Service provisioning tasks were successfully executed from end to end with low routing overhead, confirming the feasibility of interoperable and scalable MAS for optical network automation.

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References

- [1] C. Sun, X. Yang, N. Di Cicco, R. Ayassi, V. Garbhapu, P. Stavrou, M. Tornatore, G. Charlet, and Y. Pointurier, "Experimental demonstration of local AI-Agents for lifecycle management and control automation of optical networks," *Journal of Optical Communications and Networking*, vol. 17, no. 8, pp. C82–C92, 2025. DOI: [10.1364/JOCN.550286](https://doi.org/10.1364/JOCN.550286).
- [2] C. Natalino, A. Panahi, N. Mohammadiha, and P. Monti, "AI/ML-as-a-Service for Optical Network Automation: Use Cases and Challenges [Invited]," *Journal of Optical Communications and Networking*, vol. 16, no. 2, pp. A169–A179, 2024. DOI: [10.1364/JOCN.500706](https://doi.org/10.1364/JOCN.500706).
- [3] Z. Wang, M. Han, W. Cheng, Y. Huang, P. Ji, D. Wu, M. Safari, F. Holtorf, K. AlQubaisi, N. Linke, D. Zhuo, Y. Chen, T. Wang, D. Englund, and T. Chen, "Agentic AI for Scalable and Robust Optical Systems Control," *arXiv preprint arXiv:2602.20144*, 2026. DOI: [10.48550/arXiv.2602.20144](https://doi.org/10.48550/arXiv.2602.20144).
- [4] Y. Zhang, Q. Qiu, X. Liu, X. Yu, D. Fu, X. Liu, Z. Wang, H. Lin, Y. Chen, L. Yi, W. Hu, and Q. Zhuge, "AI agent for autonomous optical networks: architectures, technologies, and prospects [Invited Tutorial]," *Journal of Optical Communications and Networking*, vol. 18, no. 2, pp. A159–A178, 2026. DOI: [10.1364/JOCN.576017](https://doi.org/10.1364/JOCN.576017).
- [5] Y. Hao, M. Hemmati, M. Vaezi, Y. You, and C. Janz, "Multi-Agent LLM-powered AI for Autonomous Optical Power Commissioning of OMS Links," in *European Conference on Optical Communications*, Copenhagen, Denmark, 2025. DOI: [10.1109/ECOC66593.2025.11263012](https://doi.org/10.1109/ECOC66593.2025.11263012).
- [6] S. Xiang, S. Wu, X. Liu, Q. Qiu, Y. Zhang, Y. Chen, L. Yi, W. Hu, and Q. Zhuge, "OptiMA: Collaborative Multi-Agent Framework for Modelling and Controlling Raman Amplifier in Intelligent Optical Networks," in *European Conference on Optical Communications*, Copenhagen, Denmark, 2025. DOI: [10.1109/ECOC66593.2025.11263259](https://doi.org/10.1109/ECOC66593.2025.11263259).
- [7] Y. Zhang, Y. Song, Y. Pang, S. Li, X. Jiang, Y. Wang, J. Li, M. Zhang, and D. Wang, "Design and Evaluation of an LLM-Based Agent for QoT Estimation and Performance Optimization in Optical Networks," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 7470–7484, 2025. DOI: [10.1109/OJCOMS.2025.3608290](https://doi.org/10.1109/OJCOMS.2025.3608290).
- [8] Y. Li, S. Yin, G. Jin, X. Liu, M. Cai, and S. Huang, "LLM-Based Multi-Agent Framework for Automated Optical Network Optimization and Planning," in *Asia Communications and Photonics Conference*, Suzhou, China, 2025. DOI: [10.1109/ACP66871.2025.11350739](https://doi.org/10.1109/ACP66871.2025.11350739).
- [9] Y. Zhang, Y. Song, S. Li, Y. Shi, S. Shen, X. Tang, M. Zhang, and D. Wang, "Generative AI-Driven Hierarchical Multi-Agent Framework for Zero-Touch Optical Networks," *IEEE Communications Magazine*, January 2026, pp. 24–31. DOI: [10.1109/MCOM.001.2500117](https://doi.org/10.1109/MCOM.001.2500117).
- [10] G. Borraccini, S. Straullu, A. Giorgetti, R. Ambrosone, E. Virgillito, A. D'Amico, R. D'Ingillo, F. Aquilino, A. Nespola, N. Sambo, F. Cugini, and V. Curri, "Experimental Demonstration of Partially Disaggregated Optical Network Control Using the Physical Layer Digital Twin," *IEEE Transactions on Network and Service Management*, vol. 20, no. 3, pp. 2343–2355, 2023. DOI: [10.1109/TNSM.2023.3288823](https://doi.org/10.1109/TNSM.2023.3288823).
- [11] X. Xu, H. Chen, R. Ryf, S. Bidkar, J. Simsarian, J. McLaird, P. Rea, N. Fontaine, M. Mazur, L. Dallachiesa, and D. Neilson, "Cross-Domain Orchestration with Multi-Agent LLM Framework for Enhanced Task Automation," in *Optical Fiber Communication Conference (OFC) 2025*, San Francisco, California, United States, 2025, paper M3Z.10.
- [12] S. Xu, K. Ishii, N. Yoshikane, T. Tsuritani, Y. Awaji, and S. Namiki, "Integration and Control of Heterogeneous Telecom and Data Center Optical Networks Aided by FBD and TAPI for Enhancing Large-scale Optical Path Services and Network Resiliency," in *2021 European Conference on Optical Communication*, Bordeaux, France, 2021, pp. 1–4. DOI: [10.1109/ECOC52684.2021.9606081](https://doi.org/10.1109/ECOC52684.2021.9606081).
- [13] A2A Protocol Community, "A2A Protocol - Agent2Agent Communication," 2025. Available: <https://a2aproTOCOL.ai/>. Accessed on 20 April 2026.
- [14] D. Lopez, N. Romo Moreno, L. Tailhardat, Q. Ma, Q. Wu, Y. Yang, and S. Prabhu, "Applicability of A2A to the Network Management," Internet-Draft draft-yang-nmrg-a2a-nm-02, Internet Engineering Task Force, 26 February 2026. Available: <https://datatracker.ietf.org/doc/draft-yang-nmrg-a2a-nm-02/>. Accessed on 20 April 2026.
- [15] J. Yan and X. Zhang, "Applicability of A2A Protocol for Network Management Agents," Internet-Draft draft-yan-a2a-device-agent-applicability-01, Internet Engineering Task Force, 11 February 2026. Available: <https://datatracker.ietf.org/doc/draft-yan-a2a-device-agent-applicability/01/>. Accessed on 20 April 2026.
- [16] TM Forum, "IG1453 Agent to Agent Protocol for Telecoms (A2A-T) v2.0.0," 23 January 2026. Available: <https://www.tmforum.org/resources/introductory-guide/ig1453-agent-to-agent-protocol-for-telecoms-a2a-t-v2-0-0/>. Accessed on 20 April 2026.
- [17] Agent2Agent Protocol Community, "AgentCard," Agent2Agent Protocol Documentation. Available: <https://agent2agent.info/docs/concepts/agentcard/>. Accessed on 20 April 2026.
- [18] Agent Development Kit (ADK), "Agent Development Kit (ADK)," Available: <https://adk.dev/>. Accessed on 20 April 2026.
- [19] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, et. al "Qwen3 Technical Report," *arXiv preprint arXiv:2505.09388*, 2025. DOI: [10.48550/arXiv.2505.09388](https://doi.org/10.48550/arXiv.2505.09388).
- [20] A. Ferrari, M. Filer, K. Balasubramanian, Y. Yin, E. Le Rouzic, J. Kandrát, G. Grammel, G. Galimberti, and V. Curri, "GNPy: an open source application for physical layer aware open optical networks," *Journal of Optical Communications and Networking*, vol. 12, no. 6, pp. C31–C40, 2020. DOI: [10.1364/JOCN.382906](https://doi.org/10.1364/JOCN.382906).